

Artificial Intelligence Internship Project

Project 3: Smart Face Recognition Attendance System

This project is an AI-based attendance system that uses facial recognition technology to automatically detect and recognize students or employees using a webcam. The system marks attendance in real-time and stores records in CSV or database format.

Main Features

- Real-time face detection using webcam
- Automatic attendance marking
- Stores attendance with date and time
- Voice greeting system for recognized users
- Unknown person detection and alert
- Blink detection for security
- Modern GUI dashboard
- Student registration panel
- Attendance analytics and reports
- CSV / Excel attendance export

Technologies Used

- Python
- OpenCV
- Face Recognition Library
- NumPy
- Pandas
- Tkinter / PyQt
- SQLite / CSV
- Text-to-Speech Engine

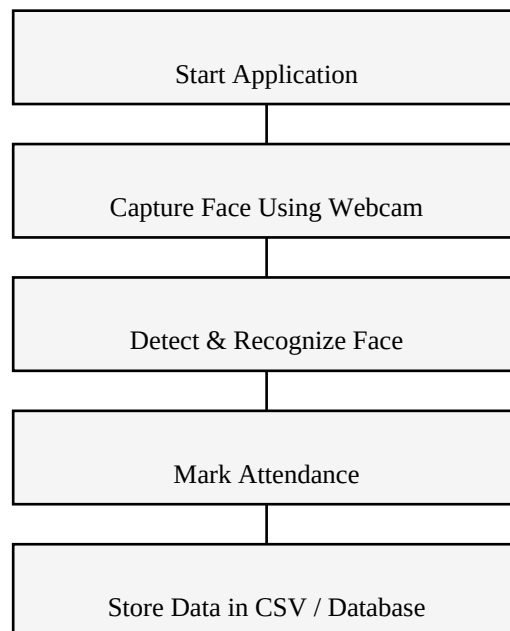
Working of the System

- User opens the attendance application.
- Webcam starts capturing live video.
- AI detects human faces from video frames.
- Detected faces are matched with stored face data.
- If face is recognized, attendance is marked automatically.
- System stores name, date, and time in attendance file.
- If face is unknown, alert message is shown.

Concepts & Skills Learned

- Artificial Intelligence
- Computer Vision
- Face Detection
- Deep Learning Basics
- Image Processing
- Real-Time Video Processing
- GUI Development
- Database Management

System Model / Flow Diagram



Advantages of the Project

- Reduces manual attendance work
- Saves time
- Improves accuracy
- Provides secure attendance system
- Useful for colleges and offices
- Prevents proxy attendance

Future Enhancements

- Cloud database integration
- Mobile application support
- Mask detection
- Online attendance portal
- Email attendance reports
- AI analytics dashboard

Conclusion:

The Smart Face Recognition Attendance System is an advanced AI project that combines computer vision and automation to create a secure and efficient attendance system. This project helps in understanding practical implementation of AI in real-world applications.